KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Quadratic and cubic expressions

Substitute a linear expression into quadratic, cubic and product terms.

10 MINUTES / 4 QUESTIONS / AS-08

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $y = 3x - 2$

Write in terms of x and simplify

Simplify $2x^3 - 2x^2 + 5y + 1$

2 Given $y = 5x - 2$

Write in terms of x and simplify

Simplify $6x^2 + 4y + 6$

3 Given $y = 3x$

Write in terms of x and simplify

Simplify $3x^2 + 5xy + 5y - 4$

4 Given $y = 3x$

Write in terms of x and simplify

Simplify $4x^2 + 3xy + 3y + 2$

Answer sheet

Quadratic and cubic expressions | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $2x^3 - 2x^2 + 15x - 9$

02 $6x^2 + 20x - 2$

03 $18x^2 + 15x - 4$

04 $13x^2 + 9x + 2$

Worked solutions

Quadratic and cubic expressions

01 Given $y = 3x - 2$

$$2x^3 - 2x^2 + 5y + 1$$

Substitute $y = 3x - 2$.

$$2x^3 - 2x^2 + 5(3x - 2) + 1$$

$$2x^3 - 2x^2 + 15x - 10 + 1$$

$$2x^3 - 2x^2 + 15x - 9$$

02 Given $y = 5x - 2$

$$6x^2 + 4y + 6$$

Substitute $y = 5x - 2$.

$$6x^2 + 4(5x - 2) + 6$$

$$6x^2 + 20x - 8 + 6$$

$$6x^2 + 20x - 2$$

Worked solutions

Quadratic and cubic expressions

03 Given $y = 3x$

$$3x^2 + 5xy + 5y - 4$$

Substitute $y = 3x$.

$$3x^2 + 5x(3x) + 5(3x) - 4$$

$$3x^2 + 15x^2 + 15x - 4$$

$$(3 + 15)x^2 + 15x - 4$$

$$18x^2 + 15x - 4$$

04 Given $y = 3x$

$$4x^2 + 3xy + 3y + 2$$

Substitute $y = 3x$.

$$4x^2 + 3x(3x) + 3(3x) + 2$$

$$4x^2 + 9x^2 + 9x + 2$$

$$(4 + 9)x^2 + 9x + 2$$

$$13x^2 + 9x + 2$$